

Aleksander Szymczak, PhD

Microbiome researcher, astrobiology enthusiast, Post-doctoral Researcher in [OMRF](#)

 szymczakaleksander@gmail.com

[LinkedIn](#) [ResearchGate](#) [Github](#) [Download Resume in PDF](#)

Tech stack

Python, R, Data Science

Bash, Powershell

Next-Generation Sequencing

Illumina, Ion Torrent

Microbial & Cell culturing

Linux, macOS, Windows

Education

PhD Studies

Hirszfeld Institute of Immunology and Experimental Therapy, Wroclaw, Poland

10/2018 - 05/2023

"Viral and bacterial components of the human microbiome: profiling in selected gastrointestinal disease states"

- Laboratory of Phage Molecular Biology, Supervisor: Prof. Krystyna Dąbrowska

Master of Science

University of Lodz, Faculty of Biology and Environmental protection, Lodz, Poland

10/2016 - 07/2018

"Application of the next generation sequencing in the analysis of the bactericidal agent's effect on microbiota"

- Department of Molecular Microbiology, Supervisor: Prof. Paweł Stączek
-

Bachelor of Science

University of Lodz, Faculty of Biology and Environmental protection, Lodz, Poland

10/2013 - 09/2016

"Strategies for bacterial survival in urinary tract infections"

- Department of Bacterial Biology, Supervisor: Prof. Agnieszka Torzewska

Work History

Principal Investigator - K99/R00

Oklahoma Medical Research Foundation

08/2025 - present

Articular Microbiome for Prediction and Potential Therapy of Periprosthetic Joint Infections

Leverage bacterial and viral community data identified to develop phage antimicrobial therapy to reduce PJI risk and test in vitro

- Award number: 1K99AR086336-01

Senior Postdoctoral Scientist

Oklahoma Medical Research Foundation

06/2023 - present

Investigating the role of the microbiome in osteoarthritis, focusing on how microbial communities within human and mouse cartilage change as the disease progresses

Exploring interventions that modify the microbiome to potentially prevent or reduce the severity of osteoarthritis by addressing changes associated with various risk factors

- Matlock Jeffries' Laboratory

Research Scholar

Oklahoma Medical Research Foundation, BioLAB program administered by Fulbright Comission

06/2022 - 05/2023

Role of autophagy in Sjorgen's syndrome and Systemic Lupus Erythematosus (SLE)

- Christopher J. Lessard's Laboratory
-

Junior Researcher

Genomtec SA, Wroclaw, Poland

10/2020 - 06/2022

BLAST analysis, Loop Mediated Isothermal Amplification (LAMP) and Real Time PCR molecular pathogen tests design and implementation

- Part of the team developing a mobile platform for genetic diagnostics at the point of patient care
-

Research Student Visitor

Stanford University School of Medicine

09/2021 - 10/2021

Paul Bollyky's Laboratory

- Application of RNAseq in bacteriophage research
-

Principal Investigator

PRELUDIUM 16 - National Science Center grant

08/2019 - 06/2022

“Systemic translocation of bacteriophages as a part of human intestine microbiome”

Budget: \$55000 ~ 210 000 PLN

Grant number: 2018/31/N/NZ6/02584

- Illumina metagenomic shotgun sequencing, Real-Time PCR, Isolation of metagenomic DNA, ultracentrifugation
-

Laboratory Assistant

Research and Development Center in Regional Specialist Hospital in Wrocław

09/2019 - 10.2021

16S rRNA sequencing, Ion Torrent system maintenance, COVID-19 protein research

- Operating Ion Torrent PGM and Ion Torrent S5 Next Next-Generation Sequencing systems , library preparation and quantification
-

Intern

Mobility Programme of Undergraduate Students and PhD students - MOST

10.2017 - 09.2018

University of Wrocław

- Two semesters at Faculty of Biological Sciences and Faculty of Biotechnology, University of Wrocław, Poland
-

Laboratory Technician

Łukasiewicz Research Network - PORT Polish Center for Technology Development - Clean Room laboratory

2017 - 2018

Clean Room laboratory

- Maintenance of ppm standards in the air in laboratory as well as microbiological purity in the clean room area

Projects & Accomplishments

 K99/R00 NIAMS grant "Articular Microbiome for Prediction and Potential Therapy of Periprosthetic Joint Infections"

 Preludium 16 grant “Systemic translocation of bacteriophages as a part of human intestine microbiome”

Publications

 Multimodal peripheral blood immunophenotyping and single cell epigenetic analysis of knee osteoarthritis reveals expansion and epigenetic dysregulation of regulatory T-cell and nonclassical monocyte populations

Aleksander Szymczak, Gabriella Dyson, Montana Barrett, Cindy Miranda Brawner, Nicholas Hanebutt, Carla Guthridge, Caleb Marlin, Joel Guthridge, Padmaja Mehta-D'souza, Matlock A. Jeffries

Arthritis & Rheumatology IF=6.54

 Peripheral blood DNA methylation changes are predictive of incident radiographic osteoarthritis development: Data and biospecimens from the osteoarthritis initiative

Cindy Miranda Brawner, Gabriella Dyson, Montana Barrett, Nicholas Hanebutt, **Aleksander Szymczak**, Padmaja Mehta-D'souza, Matlock A. Jeffries

Osteoarthritis and Cartilage Open IF=2.8

 Metabolic pathways profiling of aerobiomes in the Earth's lower atmosphere

Anita Brzoza, Kacper Dudczak, Michal Jan Winnicki, Krystian Jan Latos, Julia Jakiela, Barbara Szaflarska, Anna Lewkowicz, **Aleksander Szymczak**

International Journal of Astrobiology IF=1.6

 Ketogenic Diet-Associated Worsening of Osteoarthritis Histologic Severity, Increased Pain Sensitivity, and Gut Microbiome Dysbiosis in Mice

Gabby Dyson, Montana Barrett, Leoni Schlupp, Emmaline Prinz, Nicholas Hannebut, **Aleksander Szymczak**, Cindy Miranda Brawner, Matlock Jeffries

 [Preprint] Triadic Dynamics of Gastric Bacterial Microbiome, Phageome, and Host Genotype, with Implications for Disease Associations

Aleksander Szymczak, Katarzyna Gembara, Joanna Majewska, Paulina Miernikiewicz, Stanisław Ferenc, Jan Gnus, Marlena Kłak, Natalia Jędruchiewicz, Wojciech Witkiewicz, Krystyna Dąbrowska

bioRxiv.org

 [Preprint] Phageome transfer from gut to circulation and its regulation by human immunity

Aleksander Szymczak, Katarzyna Gembara, Stanisław Ferenc, Joanna Majewska, Paulina Miernikiewicz, Marek Harhala, Izabela Rybicka, Dominik Strapagiel, Marcin Słomka, Jakub Lach, Jan Gnus, Paweł Stączek, Wojciech Witkiewicz, Krystyna Dąbrowska

bioRxiv.org

 OA susceptibility in mice is partially mediated by the gut microbiome, is transferrable via microbiome transplantation and is associated with immunophenotype changes

Emmaline Prinz, Leoni Schlupp, Gabby Dyson, Montana Barret, **Aleksander Szymczak**, Cassandra Velasco, Vladislav Izda, Christopher Dunn, Matlock Jeffries

Annals of the Rheumatic Diseases IF=27.4

 Evolution of the T4 phage virion is driven by selection pressure from non-bacterial factors

Joanna Majewska, Paulina Miernikiewicz, **Aleksander Szymczak**, Zuzanna Kazmierczak, Tomasz Goszczynski, Barbara Owczarek, Izabela Rybicka, Jarosław Ciekot, Krystyna Dąbrowska

Microbiology Spectrum IF=9.043

 New Phage-Derived Antibacterial Enzyme PolaR Targeting *Rothia* spp.

Paulina Miernikiewicz, Jakub Barylski, Aleksandra Wilczak, Anna Dragos, Izabela Rybicka, Sophia Baldysz, **Aleksander Szymczak**, Iztok Dogsa, Kostiantyn Rokush, Marek Harhala, Jarosław Ciekot, Stanisław Ferenc, Jan Gnus, Wojciech Witkiewicz, Krystyna Dąbrowska,

 SARS-CoV-2 IgG Amongst Dental Workers During the COVID-19 Pandemic

Irena Dus-Ilnicka, Anna Szczypińska, Amadeusz Kuzniarski, **Aleksander Szymczak**, Lila Pawlik-Sobecka, Małgorzata Radwan-Oczko

International Dental Journal IF=2.512

 Hidden fraction of Polish population immune to SARS-CoV-2 in May 2021

Wiktoria Budziar, Katarzyna Gembara, Marek Harhala, **Aleksander Szymczak**, Natalia Jedruchniewicz, Krzysztof Baniecki, Aleksandra Pikies, Artur Nahorecki, Agnieszka Hoffmann, Amelia Kardaś, Alina Szewczyk-Dąbrowska, Tomasz Klimek, Zuzanna Kaźmierczak, Wojciech Witkiewicz, Kamil Barczyk, Krystyna Dąbrowska

PLoS One IF=3.24

 Antibodies specific to SARS-CoV-2 proteins N, S and E in COVID-19 patients in the normal population and in historical samples

Aleksander Szymczak, Natalia Jedruchniewicz, Alessandro Torelli, Agata Kaczmarzyk-Radka, Rosa Coluccio, Marlena Klak, Andrzej Konieczny, Stanisław Ferenc, Witkiewicz Wojciech, Krystyna Dąbrowska

Journal General Virology IF=4.87

 Laboratory Medicine in SARS-CoV-2 Diagnostics. Lessons Learned from a Pandemic

Dus-Ilnicka Irena, **Aleksander Szymczak**, Małgorzata Małodobra-Mazur, Miron Tokarski

Healthcare IF=3.16

 Application of 16S rRNA gene sequencing in *Helicobacter pylori* detection

Aleksander Szymczak, Stanisław Ferenc, Joanna Majewska, Paulina Miernikiewicz, Jan Gnus, Wojciech Witkiewicz, Krystyna Dąbrowska

PeerJ IF=3.06

 Safety Studies of Pneumococcal Endolysins Cpl-1 and Pal

Marek Harhala, Daniel Nelson, Paulina Miernikiewicz, Ryan Heselpoth, Beata Brzezicka, Joanna Majewska, Sara Linden, Xiaoran Shang, **Aleksander Szymczak**, Dorota Lecion, Karolina Marek-Bukowiec, Marlena Klak, Bartosz Wojciechowicz, Karolina Lahutta, Andrzej Konieczny, Krystyna Dabrowska

Viruses IF=3.81

 Induction of Phage-Specific Antibodies by Two Therapeutic Staphylococcal Bacteriophages Administered per os

Joanna Majewska, Zuzanna Kazmierczak, Karolina Lahutta, Dorota Lecion, **Aleksander Szymczak**, Paulina Miernikiewicz, Jakub Drapala, Marek Harhala, Karolina Marek-Bukowiec, Natalia Jedruchniewicz, Barbara Owczwarek, Andrzej Gorski, Krystyna Dabrowska

Frontiers in Immunology IF=8.786

Scientific Conferences

scATAC-seq identifies predictive chromatin accessibility biomarkers in peripheral blood mononuclear cells of knee osteoarthritis patients

Szymczak A, Dyson G, Barrett M, Brawner CM, Hanebutt N, Jeffries MA

Orthopaedic Research Society Annual Meeting, 2026, Paper No. 1384

OA-associated pain sensitivity is reduced in OA-protected MRL/MpJ mice, partially mediated by the gut microbiome, and transferable via microbiome transplantation

Dyson G, Barrett M, Hanebutt N, **Szymczak A**, Brawner CM, Dube P, Cherian E, Lopez A, Mehta-D'souza P, Jeffries MA

Orthopaedic Research Society Annual Meeting, 2026, Paper No. 338

Heterochronic gut microbiome transplantation augments the risk of osteoarthritis in mice

Hanebutt N, Barrett M, Dyson G, **Szymczak A**, Brawner CM, Cherian E, Dube P, Lopez A, Mehta-D'souza P, Jeffries MA

Orthopaedic Research Society Annual Meeting, 2026, Paper No. 199

Bacteriophage Communities in Osteoarthritic Joints

Szymczak A, Brawner C, Hanebutt N, Dyson G, Barrett M, Dube P, Cherian E, Mehta-D'souza P, Jeffries M

Orthopaedic Research Society Annual Meeting, 2026, Paper No. 1884

Specific gut microbiome remodeling occurs after induction of OA via DMM surgery compared to sham surgery

Szymczak A, Miranda C, Hanebutt N, Dyson G, Barrett M, Jeffries MA

American College of Rheumatology Convergence, 2025

OA protection in African spiny (Acomys) mice is partially mediated by the gut microbiome and is transferrable to wild-type mice via microbiome transplantation

Dyson G, Barrett M, Hanebutt N, **Szymczak A**, Brawner C, Dube P, Mehta-D'souza P, Seifert A, Jeffries MA

American College of Rheumatology Convergence, 2025

Sex differences in knee joint microbial DNA of C57BL/6 mice and the effect of cross-sex microbiome transplantation

Kurra VV, Lopez A, **Szymczak A**, Dyson G, Dube P, Mehta-D'souza P, Jeffries M

American College of Rheumatology Convergence, 2025

OA-associated pain sensitivity is reduced in OA-protected MRL/MpJ mice, partially mediated by the gut microbiome, and transferable via microbiome transplantation

Dyson G, Barrett M, Hanebutt N, **Szymczak A**, Brawner C, Dube P, Lopez A, Mehta-D'souza P, Jeffries MA

American College of Rheumatology Convergence, 2025

Bacteriophage Communities in Osteoarthritis-Associated Microbiomes

Szymczak A, Brawner C, Dyson G, Barrett M, Hanebutt N, Dube P, Mehta-D'souza P, Jeffries M

From Therapeutic Agents to Disease Drivers: AI-Powered Discovery of Bacteriophage Roles in Osteoarthritis

Szymczak A, Brawner C, Dyson G, Barrett M, Hanebutt N, Dube P, Mehta-D'souza P, Jeffries M

Microbial Dark Matter, 2025

The Phages of Arthritis: The Role of the Virome in Osteoarthritis

Szymczak A

Cartilage Repair Strategies to Alleviate Arthritis Pain (CARE-AP) Collaborative Research Meeting, 2025

Diverse bacteriophage communities in osteoarthritis patients

Szymczak A, Miranda C, Hanebutt N, Dyson G, Barret M, Jeffries M

Evergreen phage meeting, 2025

Preclinical knee OA can be defined by peripheral blood epigenetic patterns

Miranda C, Hanebutt N, Dyson G, Barrett M, **Szymczak A**, Jeffries MA

Orthopaedic Research Society Annual Meeting, 2025

Deep immunophenotyping of knee osteoarthritis patients reveals alterations in T-cell, Regulatory-T, and Monocyte subpopulations

Barrett M, Dyson G, Hannebut N, **Szymczak A**, Brawner C, Jeffries MA

Orthopaedic Research Society Annual Meeting, 2025

Computational Exploration of Unknown Sequences in Earth's Atmospheric Microbiomes

Latos K, Brzoza A, Jakięła J, Winnicki M, Dudczak K, Szaflarska B, Lewkowicz A, **Szymczak A**

Metabolic profiling of aerobiomes in the Earth's troposphere

Lewkowicz A, Jakiela J, Winnicki M, Brzoza A, Dudczak K, Latos K, Szaflarska B, **Szymczak A**

European Astrobiology Network Association EANA, 2024

Microbial genomes' integrity during space missions

Dudczak K, Brzoza A, Jakiela J, Latos K, Winnicki M, Szaflarska B, Lewkowicz A, **Szymczak A**

Life and space, 2024

The development of future radiographic knee osteoarthritis can be predicted by peripheral blood epigenetic models

Miranda C, Hanebutt N, Dyson G, Barrett M, **Szymczak A**, Jeffries M

American College of Rheumatology Convergence Annual Meeting, 2024

Deep immunophenotyping of knee osteoarthritis patients reveals alterations in T cell, Treg, and monocyte subpopulations

Montana B, Dyson G, Hanebutt N, **Szymczak A**, Miranda C, Jeffries M

American College of Rheumatology Convergence Annual Meeting, 2024

Exploring Earth's Aerobiome through Similarity Search Analysis

Winnicki M, Jakiela J, Brzoza A, Dudczak K, Latos K, Lewkowicz A, Szaflarska B, **Szymczak A**

2024 Galaxy Community Conference (GCC2024), 2024

Baseline peripheral blood epigenetic age is accelerated among individuals with knee osteoarthritis who experience dual radiographic+pain and radiographic-only progression but not pain-only progression

Szymczak A, Miranda C, Hanebutt N, Dyson G, Barrett M, Arbeeva L, Golightly Y, Nelson A, Jeffries M

OARSI World congress of Osteoarthritis, 2024

The gut microbiome in mice is altered during osteoarthritis development following destabilization of the medial meniscus (DMM) but not following sham surgery

Szymczak A, Miranda C, Hanebutt N, Dyson G, Barrett M, Jeffries M

OARSI World congress of Osteoarthritis, 2024

Preclinical OA can be defined by peripheral blood epigenetic models

Miranda C, Hanebutt N, Dyson G, Barrett M, **Szymczak A**, Jeffries M

OARSI World congress of Osteoarthritis, 2024

Future osteoarthritis development can be predicted by peripheral blood epigenetic machine-learning models up to 96 months in advance

Miranda C, Hanebutt N, Dyson G, Barrett M, **Szymczak A**, Jeffries M

Military Health System Research Symposium (MHSRS) Annual Meeting, 2024

Preclinical OA can be defined by peripheral blood epigenetic models

Miranda M, Hanebutt N, Dyson G, Barrett M, **Szymczak A**, Jeffries M

ACR Convergence, 2024

Decoding Earth's Aerobiomes: Implications for Astrobiology and the Search for Extraterrestrial Life

Brzoza A, Dudczak K, Jakiela J, Kuska M, Szaflarska B, Naharnowicz M, Czarnota P, Winnicki M, Latos K, **Szymczak A**

2023 Galaxy Community Conference (GCC2023), 2023

Analysing Microbial Metabolic Pathways in Earth's Atmosphere: Insights for Astrobiological Research

Evolution of the T4 phage virion is driven by selection pressure from the gastrointestinal tract environment

Majewska J, Miernikiewicz P, **Szymczak A**, Kaźmierczak Z , Dabrowska K

Viruses of Microbes 2022, Abstract book p. 191

Transfer between ecological niches within human body: how gut phageome expands to serum phageome

Szymczak A, Gnus J, Ferenc S, Miernikiewicz P, Majewska J, Gembara K, Strapagiel D, Stączek P, Dąbrowska K

Viruses of Microbes 2022, Abstract book p. 191

Biofilm the 9th Passenger of Nostromo - Targeting amyloids to prevent biofouling in space,

Kuska M, Łopucki K, Okoń R, Pilska-Piotrowska M, Latos K, Kowalczyk A, **Szymczak A**, Górecki I, Janiec J, Zajkowski T, Rothschild L

Life and Space Conference 2021, Abstract Book p. 52, 29.09 - 1.10 2021

Correlation between SNPs in human genomes and bacterial and viral complexity of human microbiome

Szymczak A, Gembara K, Ferenc S, Strapagiel D, Majewska J, Miernikiewicz P, Lach, J, Gnus J, Stączek P, Witkiewicz W, Dąbrowska K

EMBL Symposium: New Approaches and Concepts in Microbiology, 7-9.06.2021, Online Abstract Book

Comparison of human stomach microbiome and virome fractions

Szymczak A, Ferenc S, Miernikiewicz P, Majewska J, Gnus J, Witkiewicz W, Dąbrowska K

Oxford Bacteriophage Conference – Phages 2019, 11-12.09.2019, Oxford UK, Book of Web p. 36

Microbiome comparison of *Helicobacter pylori* infected and non-infected patients

Gut microbiome composition in mice after 100 days of oral treatment with staphylococcal bacteriophage A3R

Szymczak A, Majewska J, Marek-Bukowiec K, Lecion D, Miernikiewicz P, Kęska W, Kołodziej M, Górski A, Dąbrowska K

EMBO Viruses of Microbes 2018, 9-13.07.2018, Wrocław, Abstract Book, p.389 0

Training Courses

R Programming" Course Certificate (2020)

Getting and Cleaning Data in R" Course Certificate (2020)

Reproducible Research" Course Certificate (2020)

The Data Scientist's Toolbox" Course Certificate (2020)

Exploratory Data Analysis" Course Certificate (2020)

Omics Data Science - Bioinformatics and Biomedical Big Data Analysis (2019-2020)

Introduction to NGS data analysis, Ideas4Biology (2019)

Good Laboratory Practice Principles and Procedures. Requirements for personnel in an Accredited Laboratory according to the PN-EN ISO 17025 (2015)

The University of North Carolina at Chapel Hill's open online course: Epidemiology:

The Basic Science of Public Health (2013)

Social activities

Oklahoma City Astronomy Club

Polish Astrobiology Society

Space Generation Advisory Council

International Association for the Exchange of Students for Technical Experience (IAESTE) Member